

Samuel Crabbendam

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Education

University of Florida, Wertheim College of Engineering

May 2028

Bachelor of Science in Aerospace Engineering

GPA: 3.6

Relevant Coursework: *Design & Manufacturing, Thermodynamics, Aerodynamics, Mechanics of Materials, Dynamics*

Skills

- Proficiency in SolidWorks, Fusion360, MATLAB, Flow5, UAS systems, Microsoft Office, 3D Print, Github

Certificates

- CSWA Certified, *SolidWorks* Gainesville, FL | 2025 - Present
- Part 107 Remote Pilot License, *West Orange High School* Winter Garden, FL | 2023 - Present
- Part 61 Private Pilot License, *Orlando Flight School, 55 hours* Orlando, FL | 2026 - Present

Experience

Payload Mechanics Engineer / Flight Dynamics Engineer | *Swamp Launch Rocketry*

Gainesville, FL | August 2024 – Present

- Designed the 2026 IREC aluminum & PLA composite 4U CubeSat system housing two graduate research experiments.
- Facilitated experiments analyzing gene expression and protein response under launch G-forces.
- Designed and assembled the 2025 IREC aluminum and ABS chase camera deployment system.
- Simulated rocket designs and stability across varying wind conditions, validated via successful launch in the Spaceport America Cup.

Fixed Wing UAS Design & Build | *Independent Project*

Gainesville, FL | July 2026 – Present

- Designing a fixed-wing reconnaissance UAS with ArduPilot GPS-guided autonomous flight, independent camera control, and a 5 ft wingspan.
- Performed aerodynamics analysis through Flow5 to validate airframe configuration and lift generation.
- Engineering a foam & PLA airframe to maximize strength-to-weight ratio, reducing mass by 40% vs prev. iteration.
- Creating a servo-actuated rotating hemispherical camera gimbal for increased visibility during flight.
- Managing \$500 project budget and 16 week fabrication timeline through iterative design-build-test methodology.

Wind Tunnel Research Assistant | *Undergraduate Research at the University of Florida*

Gainesville, FL | August 2026 – Present

- Selected to join closed loop wind tunnel research team; training on startup and safety procedures ahead of leading-edge Alula aerodynamic testing in Spring 2027.

Project Engineering Intern | *Garney Construction*

Tampa, FL | May 2026 – August 2026

- Conducted quality control inspections of structural assemblies against engineering specifications.
- Interpreted engineering drawings and specifications to generate material takeoffs and verify design compliance.
- Eliminated unnecessary sitewide production delays by creating an Excel procurement system.
- Authored technical reports, meeting documentation, and project updates while coordinating across engineering, management, vendor, and field construction teams.
- Verified material quality via standardized testing to confirm specifications were met before installation.